

Campus Event Hub

Club Events & Student Registration — Concept & Architecture Overview

1. Basic Idea & Core Features

Campus Event Hub is a centralized web platform where campus clubs create and manage events, and students discover and register for them. It replaces posters, WhatsApp groups, and paper sign-ups with a self-service digital flow, with three roles: **Students** (browse/register), **Club Coordinators** (create/manage events), and **Admins** (approve events, monitor participation).

Event Listing	Searchable, filterable feed of upcoming events (club, category, date, venue, seats left).
Registration	One-click sign-up with real-time seat tracking and duplicate-entry prevention.
Countdown Timer	Live countdown on each event page showing time left until the event starts.
QR Code Check-in	Unique QR per registration, emailed & in-app, scanned at the venue for instant attendance.
Admin Panel	Approve events, view live registrations/check-ins, export attendee lists, view analytics.

2. System Architecture

A 3-tier client-server design: React frontend(s) talk to a single REST API gateway, which routes to modular backend services, backed by a database, Redis cache, and cloud storage.

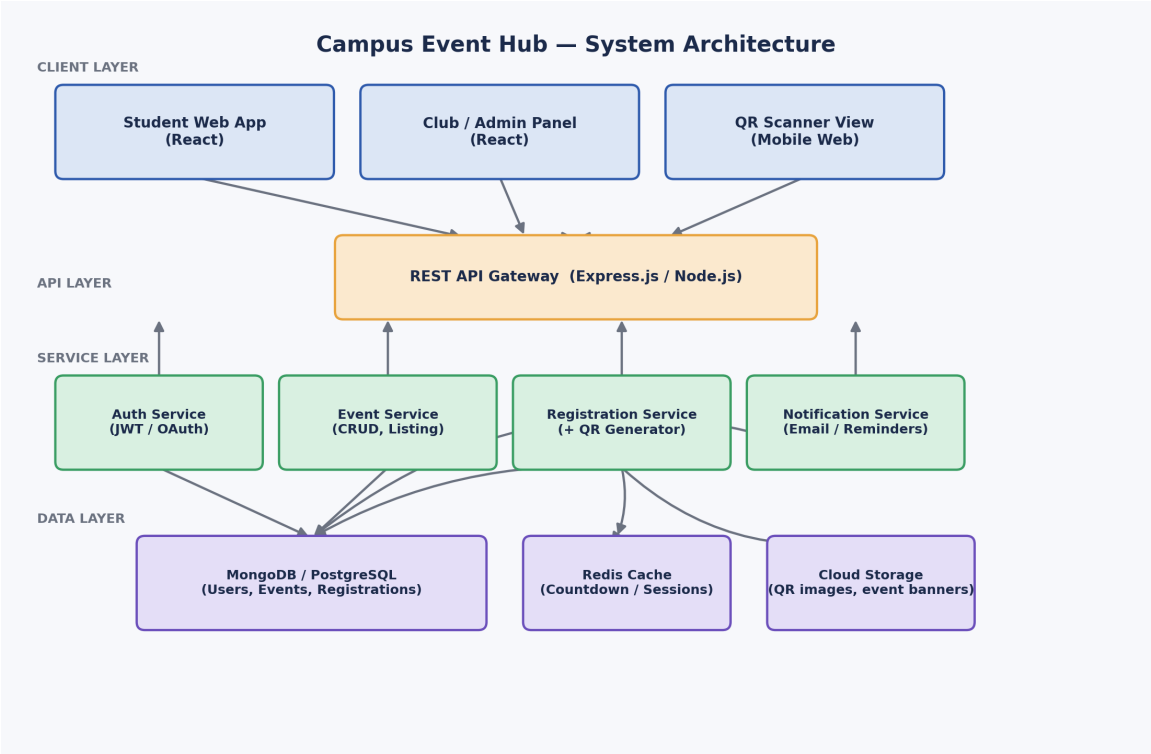


Figure 1 — High-level system architecture

3. Frontend

React.js SPA (Vite) + Tailwind CSS, React Router, Axios. Handles browsing, registration, countdown, QR display, and role-based dashboards.

- **Event Listing & Event Details** pages (with countdown)
- **Registration/Confirmation** page (shows QR)
- **Student Dashboard** — My Registrations
- **Club/Admin Panel** — create events, view registrants
- **QR Scanner View** — mobile check-in tool
- JWT-based auth with protected routes

4. Backend

Node.js + Express REST API, modular services (Auth, Event, Registration, Notification), MongoDB/PostgreSQL, Redis, cloud storage.

- **Auth Service** — signup/login, JWT, bcrypt
- **Event Service** — CRUD + listing endpoints
- **Registration Service** — register + QR generation (Node qrcode)
- **Notification Service** — email confirmations (Nodemailer/SendGrid)
- **Admin Service** — approvals + analytics endpoints

5. Architectural Workflow — Registration & QR Check-in

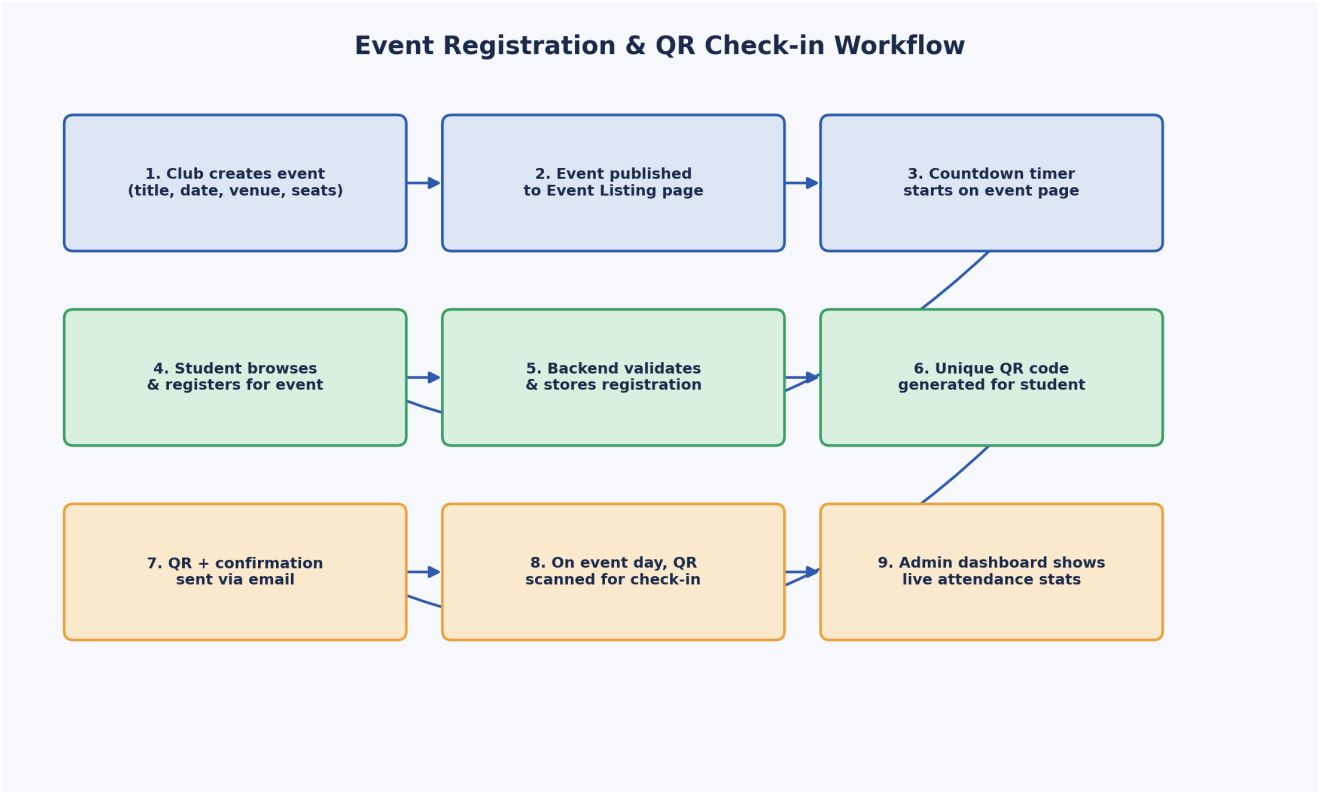


Figure 2 — End-to-end event registration and check-in flow

6. Data Model (Simplified) & Tech Stack

Entity	Key Fields
User	id, name, email, passwordHash, role
Event	id, title, clubId, venue, startTime, seatLimit, seatsFilled, status
Registration	id, userId, eventId, qrToken, status, createdAt
Club	id, name, coordinatorIds

Layer	Technology
Frontend	React.js, Tailwind CSS, React Router
Backend	Node.js, Express.js, JWT
Database	MongoDB / PostgreSQL
Cache	Redis
Storage	AWS S3 / Cloudinary
QR / Notify	qrcode.js, Nodemailer / SendGrid
Deployment	Vercel/Netlify + Render/Railway

Summary: Clubs publish events → students discover & register with a live countdown → a unique QR code is generated and emailed → QR scan at the venue marks attendance instantly → admins get real-time visibility into registrations and turnout across campus.